

Configuring UEFI settings - AMD

To enable SEV-SNP the following UEFI settings must be configured:

1. Enable AMD Secure Memory Encryption (SME):

This setting must be set to **Enabled**.

- Path: UEFI Settings → System Settings → Memory → SMEE → Enabled
- Description: This setting is required to use all Secure Encrypted Virtualization features and uses a single key to encrypt system memory and provides additional security benefits. For example, when enabled it can protect the Hypervisor and the Host OS.

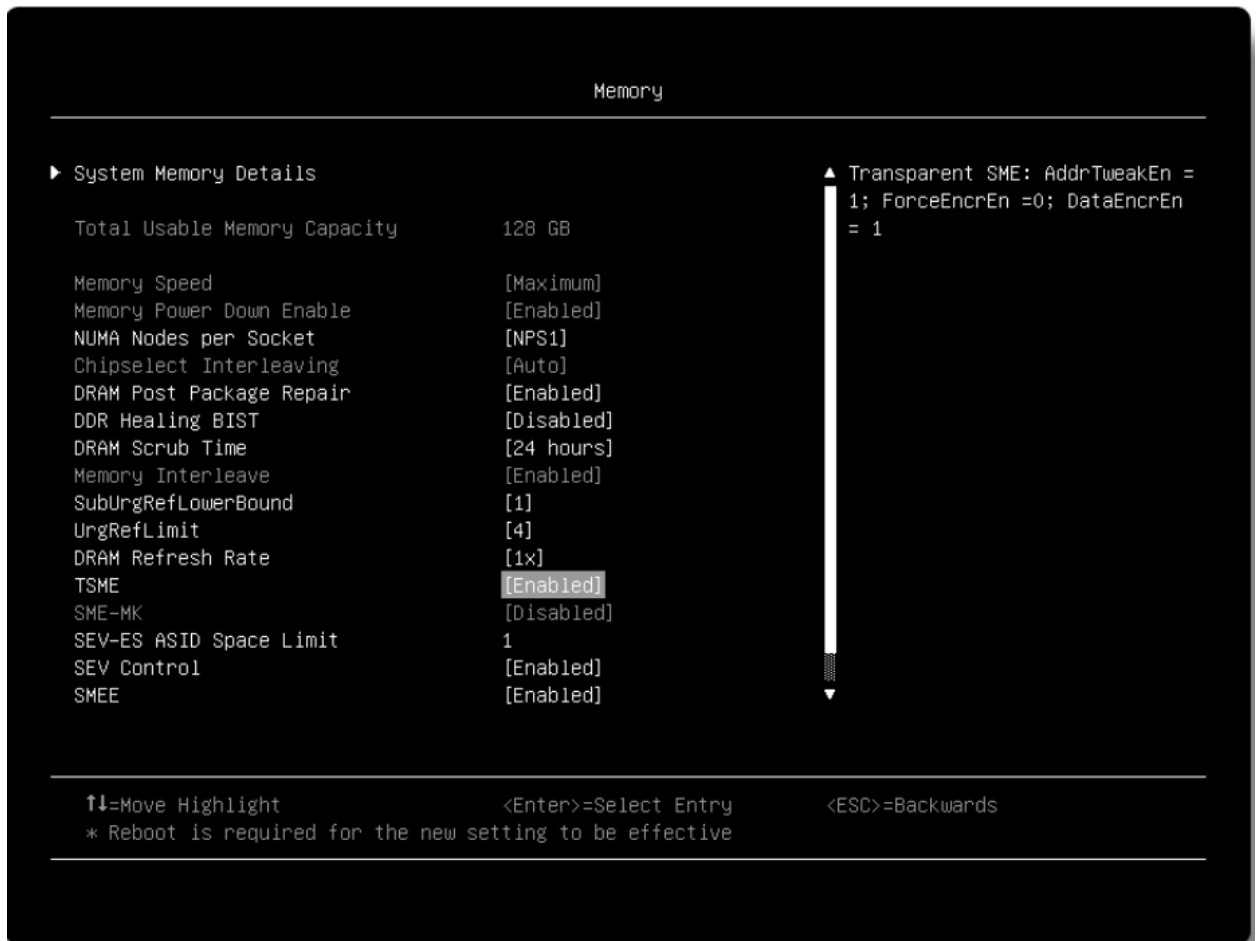


2. Enable Transparent Secure Memory Encryption (TSME):

This setting must be set to **Enabled**.

- Path: UEFI Settings → System Settings → Memory → TSME → Enabled

- Description: When enabled, it provides full memory encryption without requiring changes to the operating system or applications. It is a simplified version of AMD Secure Memory Encryption. TSME is required for DRTM.



3. Configure the Secure Encrypted Virtualization Address Space Identifier (ASID) Space Limit:

Set this value to “ASID Count +1”. SEV-ES virtual machines are the most secure virtual machines.

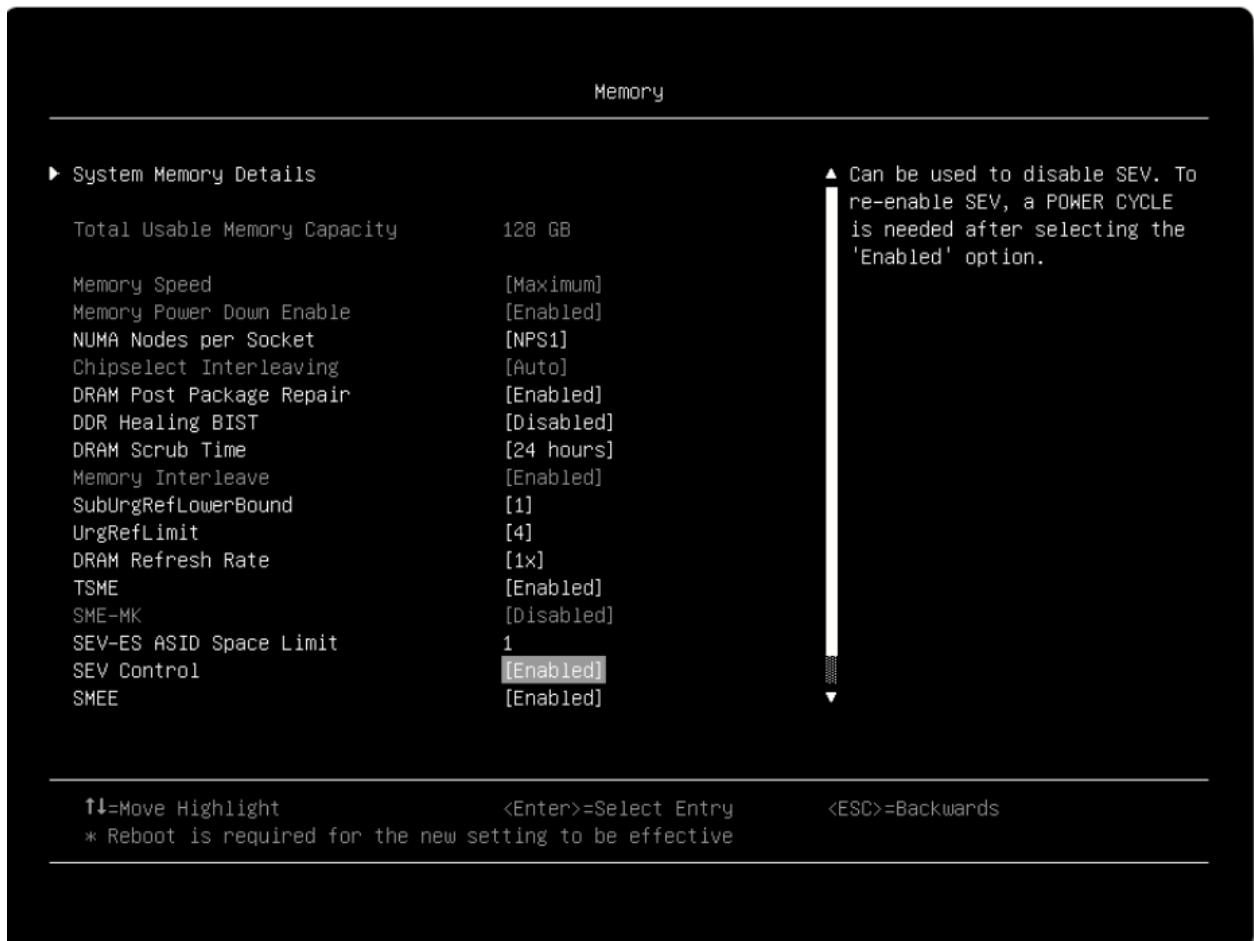
- Path: UEFI Settings → System Settings → Memory → SEV-ES ASID Space Limit → value
- Description: SEV VMs using ASIDs below the SEV-ES ASID Space Limit must enable the SEV-ES feature. ASIDs from the SEV-ES ASID Space Limit to (SEV ASID Count + 1) can only be used with SEV VMs. If this field is set to (SEV ASID Count + 1), all ASIDs are forced to be SEV-ES ASIDs.



4. Enable Secure Encrypted Virtualization Control:

This setting must be set to **Enabled**.

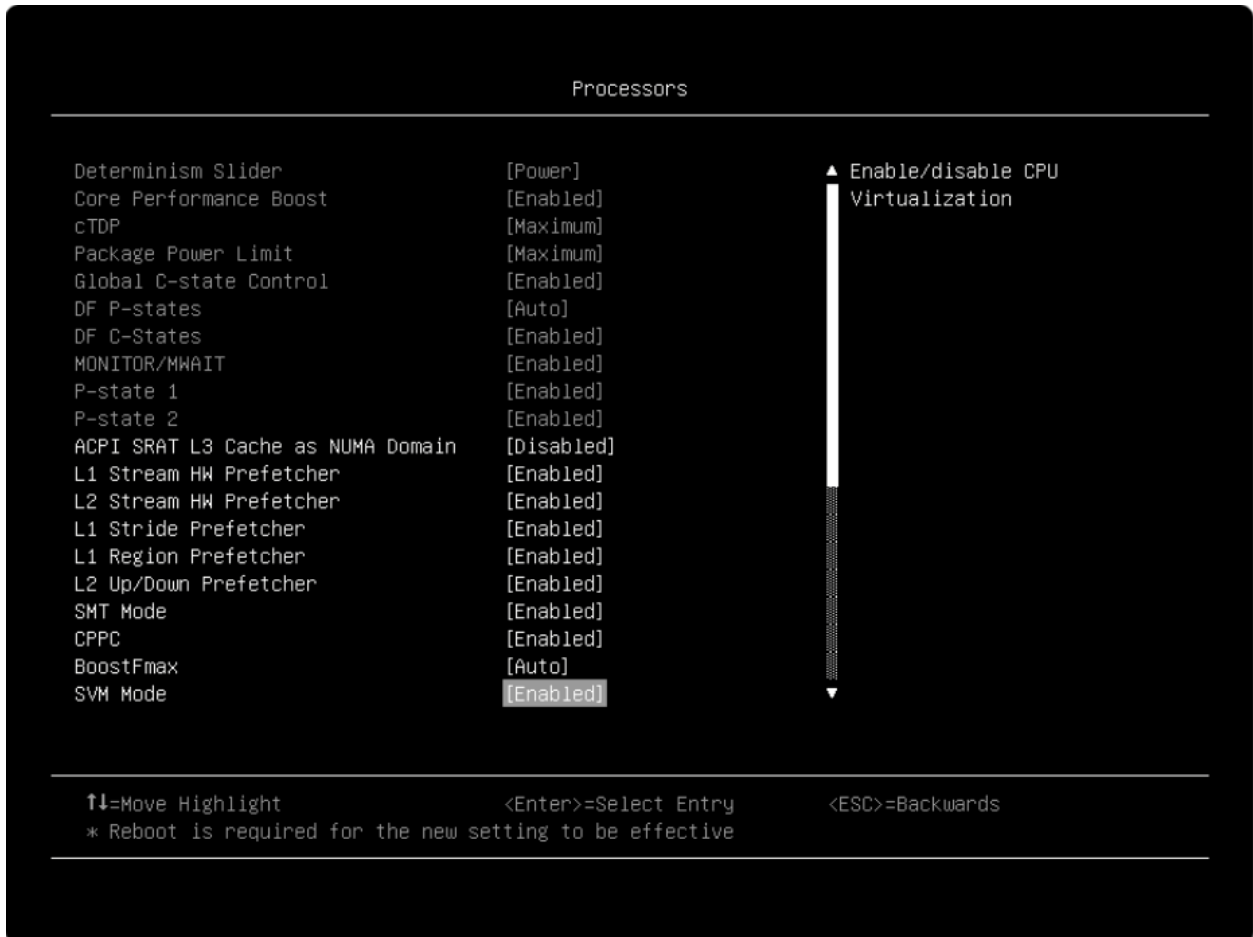
- Path: UEFI Settings → System Settings → Memory → SEV Control → Enabled
- Description: This setting enables SEV control. Enables confidential computing by protecting VM memory and state from unauthorized access.



5. Enable SVM Mode:

This setting must be set to **Enabled**.

- Path: UEFI Settings → System Settings → Processors → SVM Mode → Enabled
- Description: This setting enables CPU Virtualization. Enables Secure Virtual Machine Mode and is required for SEV-ES and SEV-SNP and for Confidential Computing features.



6. Enable Secure Encrypted Virtualization-Secure Nested Paging (SNP):

This setting must be set to **Enabled**.

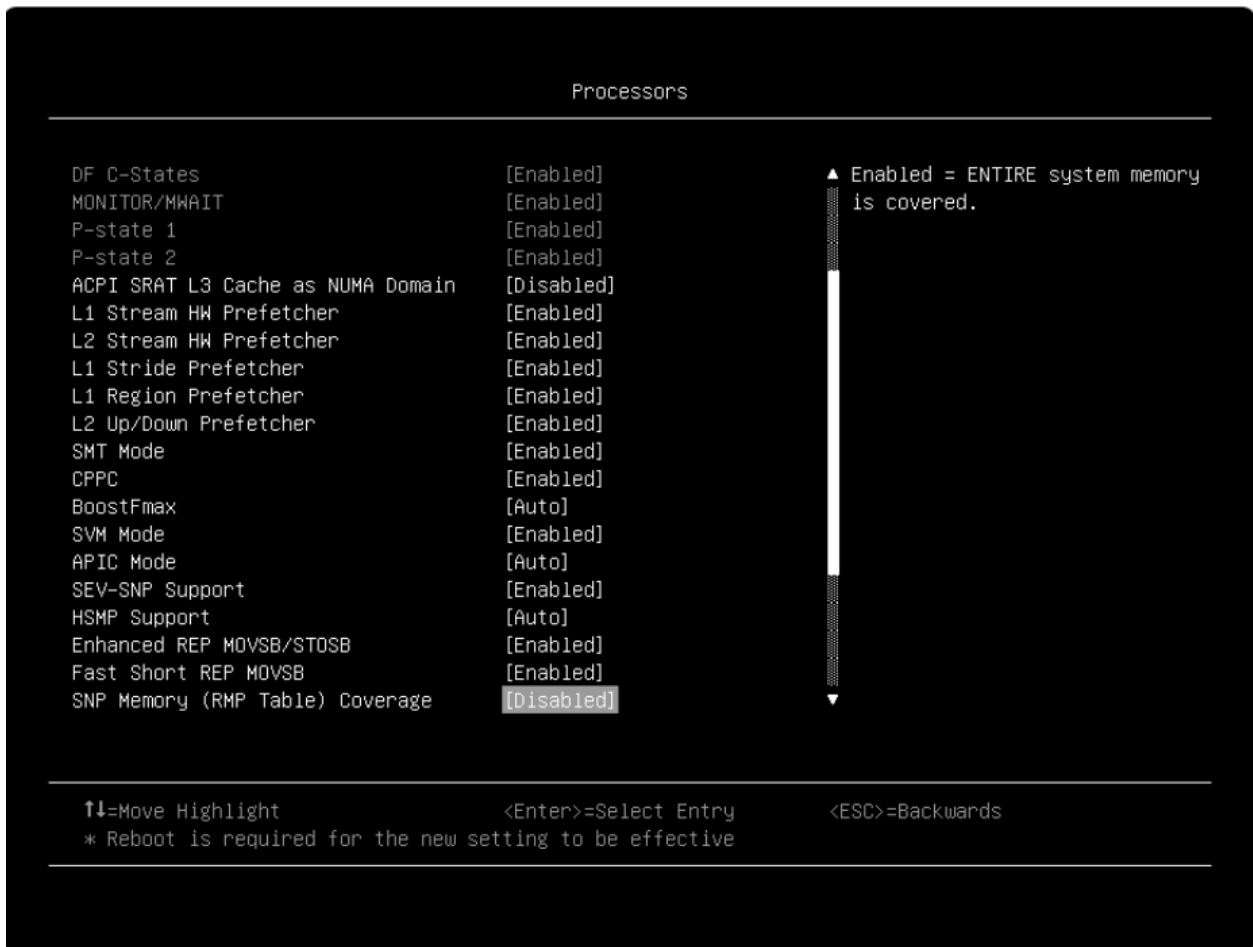
- Path: UEFI Settings → System Settings → Processors → SEV-SNP Support → Enabled
- Description: This setting enables SEV-SNP support. Provides memory integrity protection and guest isolation from the hypervisor.



7. Enable Secure Nested Paging (SNP) RMP Table Coverage:

This setting must be set to **Disabled**.

- Path: UEFI Settings → System Settings → Processors → SNP Memory (RMP Table Coverage) → Disabled
- Description: The use of the Reverse Map Table (RMP) Table to protect memory when SEV-SNP is enabled. This is not applicable to Hyper-V as it allocates and manages its own RMP table instead of platform firmware



8. Enable IOMMU:

This setting must be set to **Enabled**.

- Path: UEFI Settings → System Settings → Processors → Secured Core → IOMMU → Enabled
- Description: This setting enables Input-Output Memory Management Unit (IOMMU) support. Enables the Input-Output Memory Management Unit and is required for SEV-SNP enablement. The IOMMU is an AMD EPYC feature that manages and translates memory addresses between devices and system memory.

Secured-Core

Secured-Core	[Custom]	Enable/Disable IOMMU.
IOMMU	[Enabled]	
DMAR Support	[Disabled]	
DMA Protection	[Disabled]	
DRTM Virtual Device Support	[Disabled]	
TSME	[Enabled]	
DRTM Memory Reservation	[Disabled]	

↑↓=Move Highlight

<Enter>=Select Entry

<ESC>=Backwards

* Reboot is required for the new setting to be effective
